

1. How do you create a table with a PRIMARY KEY constraint on a single column?

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    EmployeeName VARCHAR(100)  
);
```

2. Write a SQL query to add a PRIMARY KEY constraint to an existing table.

```
ALTER TABLE Employees  
ADD PRIMARY KEY (EmployeeID);
```

3. How do you create a composite PRIMARY KEY constraint on multiple columns?

```
CREATE TABLE EmployeeDepartment (  
    EmployeeID INT,  
    DepartmentID INT,  
    PRIMARY KEY (EmployeeID, DepartmentID)  
);
```

4. Write a SQL query to add a FOREIGN KEY constraint to an existing table.

```
ALTER TABLE Employees  
ADD CONSTRAINT fk_department  
FOREIGN KEY (DepartmentID)  
REFERENCES Departments(DepartmentID);
```

5. Write a SQL query to drop a PRIMARY KEY constraint from a table.

```
ALTER TABLE Employees  
DROP PRIMARY KEY;
```

6. How do you create a UNIQUE constraint on a column in a table?

```
ALTER TABLE Employees
```

```
ADD CONSTRAINT uq_employee_name  
UNIQUE (EmployeeName);
```

7. Write a SQL query to drop a UNIQUE constraint from a table.

```
ALTER TABLE Employees  
DROP INDEX uq_employee_name;
```

8. How do you enforce the NOT NULL constraint on a column?

```
CREATE TABLE Employees (  
    EmployeeID INT,  
    EmployeeName VARCHAR(100) NOT NULL  
);
```

9. Write a SQL query to modify an existing column to enforce the NOT NULL constraint.

```
ALTER TABLE Employees  
MODIFY EmployeeName VARCHAR(100) NOT NULL;
```

10. How do you define a DEFAULT constraint for a column?

```
CREATE TABLE Employees (  
    EmployeeID INT,  
    Department VARCHAR(50) DEFAULT 'HR'  
);
```

11. Write a SQL query to alter an existing column to add a DEFAULT constraint.

```
ALTER TABLE Employees  
ALTER COLUMN Department SET DEFAULT 'HR';
```

12. How do you add a CHECK constraint to a table to ensure values fall within a specific range?

```
ALTER TABLE Employees  
ADD CONSTRAINT chk_salary  
CHECK (Salary BETWEEN 10000 AND 100000);
```

13. Write a SQL query to drop a CHECK constraint from a table.

```
ALTER TABLE Employees  
DROP CHECK chk_salary;
```

14. How do you disable all constraints on a table temporarily?

```
SET FOREIGN_KEY_CHECKS = 0;
```

15. Write a SQL query to re-enable all previously disabled constraints on a table.

```
SET FOREIGN_KEY_CHECKS = 1;
```

16. Write a SQL query to rename an existing constraint.

```
ALTER TABLE Employees  
DROP CHECK chk_salary,  
ADD CONSTRAINT chk_employee_salary  
CHECK (Salary > 10000);
```

17. How do you enforce a constraint to ensure that a column's value matches a predefined list of values?

```
ALTER TABLE Employees  
ADD CONSTRAINT chk_department  
CHECK (Department IN ('HR', 'IT', 'Finance'));
```

18. Write a SQL query to add a constraint that ensures a column's value is always greater than a certain threshold.

```
ALTER TABLE Employees  
ADD CONSTRAINT chk_salary  
CHECK (Salary > 20000);
```

19. How do you ensure that a column's value is based on a calculation involving other columns in the same row?

```
ALTER TABLE Employees
```

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```
ADD CONSTRAINT chk_total_salary  
CHECK (TotalSalary = BasicSalary + Bonus);
```